Smart Queue & Appointment Management System

CSC 2032 – Object Oriented Programming

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GitHub repository link : https://github.com/Dilki30/smart-queue-system.git

Content

1. Introduction

Queue management has become a significant challenge in hospitals, banks, government offices, and other public service organizations. Traditional manual queue systems cause long waiting times, overcrowding, and poor customer satisfaction.

The **Smart Queue Appointment System** is a web-based application developed to digitalize queue management by allowing customers to reserve appointments online, receive queue numbers, monitor queue status, and view their appointment history.

2. Problem Statement

Many organizations still rely on manual token systems.

Problems include:

- Long waiting times
- Overcrowded waiting areas
- Customers unsure about their queue position
- Time wastage
- Difficult queue management
- No online booking system
- No appointment history

3. Objectives

3.1 Main Objective

Develop an online Smart Queue Appointment System using Object-Oriented Programming concepts.

3.2 Specific Objectives

- Allow users to register and login
- Book appointments online
- Generate queue tokens automatically
- Display live queue information
- Maintain appointment history
- Improve customer experience
- Reduce waiting time

4. Proposed Solution

The proposed system provides an online platform where customers can:

- Create an account
- Login securely
- Select a service
- Book appointments
- Receive queue numbers
- View live queue status
- Access booking history
- Manage their profile

Administrators can manage queues and appointments efficiently.

5. Technologies Used

Component	**Technology**
Frontend	HTML5
Styling	CSS3
Client-side	JavaScript
Backend	Node.js
Architecture	REST API
Data Storage	JSON File Database (Can be upgraded to MySQL)
Version Control	Git & GitHub

6. System Features

6.1 User Authentication

- User Registration
- User Login
- Logout
- Session Management

6.2 Dashboard

Displays

- User information
- Upcoming appointments
- Queue statistics
- Navigation panel

6.3 Appointment Booking

Users can

- Choose service
- Select appointment date
- Select appointment time
- Submit booking

6.4 Queue Management

System automatically

- Generates queue numbers
- Displays queue position
- Shows waiting status
- Updates queue progress

6.5 Ticket Generation

After booking

- Queue number generated
- Appointment details shown
- Booking confirmation displayed

6.6 History

Users can

- View previous bookings
- View completed appointments
- Track appointment records

6.7 User Profile

Users can update

- Name
- Email
- Password
- Personal information

6.8 Admin Authentication

Admin can view queue status and cancel appointments.

7. Object-Oriented Programming Concepts

The project follows Object-Oriented Programming principles.

7.1 Encapsulation

Sensitive user information is managed securely through classes and modules.

Example:

- User class
- Ticket Handler
- Authentication Handler

7.2 Abstraction

Complex backend operations are hidden from users.

Examples:

- Queue generation
- Authentication
- Booking process

7.3 Inheritance

Future versions can include:

User
↓
Customer
↓
Administrator

Both inherit common properties from the User class.

7.4 Polymorphism

Different service handlers perform booking differently.

Examples

- Hospital Handler
- Bank Handler
- Ticket Handler

Each processes requests uniquely while using a common structure.

8. System Architecture

```
+----------------------+
|     User Browser     |
+----------+-----------+
           |
           |
     HTML/CSS/JS
           |
 REST API Requests
           |
+----------v-----------+
|      Node.js Server  |
+----------+-----------+
           |
 Business Logic
           |
+----------v-----------+
| Queue Management     |
| Authentication       |
| Ticket Generation    |
+----------+-----------+
           |
     JSON Database
```

9. Database Design

9.1 User Table

Field	Type
UserID	Integer
Name	Varchar
Email	Varchar
Password	Varchar

9.2 Appointment Table

Field	Type
AppointmentID	Integer
UserID	Integer

Field	Type
Service	Varchar
Date	Date
Time	Time
QueueNumber	Integer
Status	Varchar

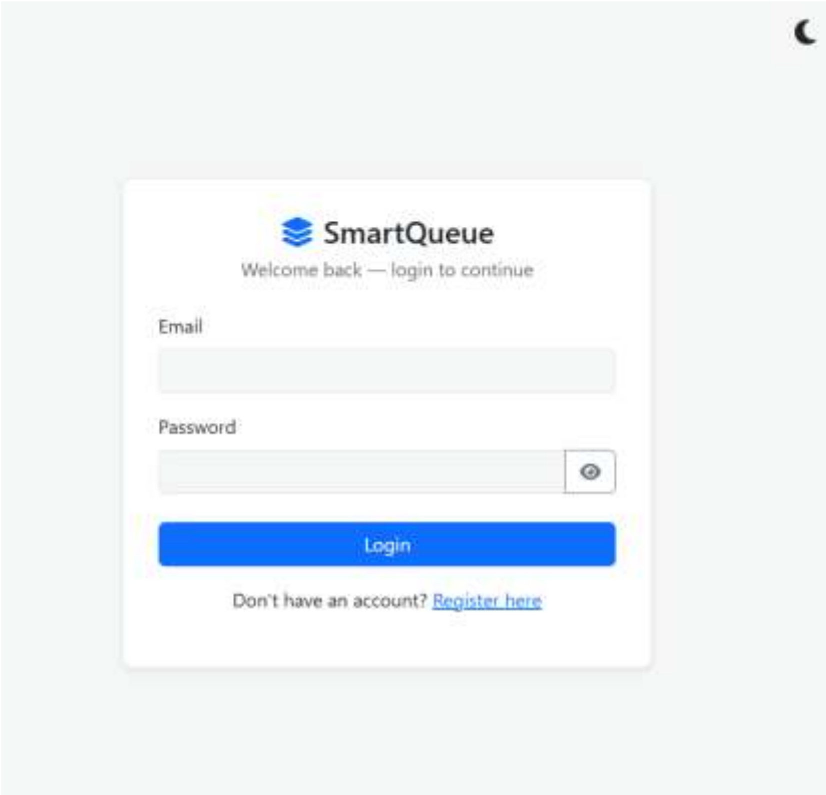
9.3 Queue Table

Field	Type
QueueID	Integer
QueueNumber	Integer
CurrentPosition	Integer
Status	Varchar

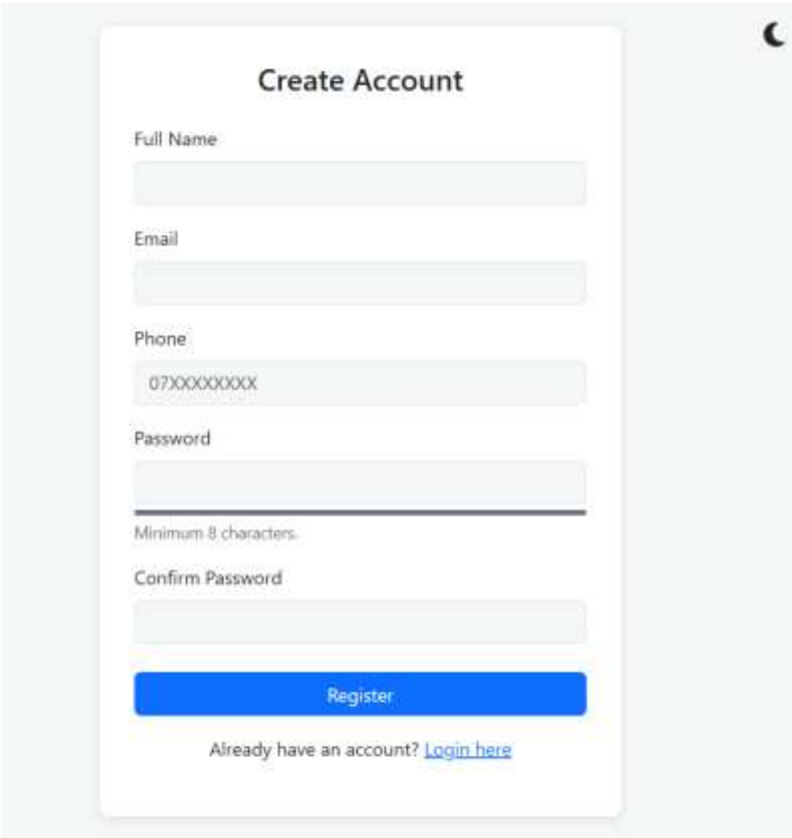
10. System Interfaces

The project contains the following pages:

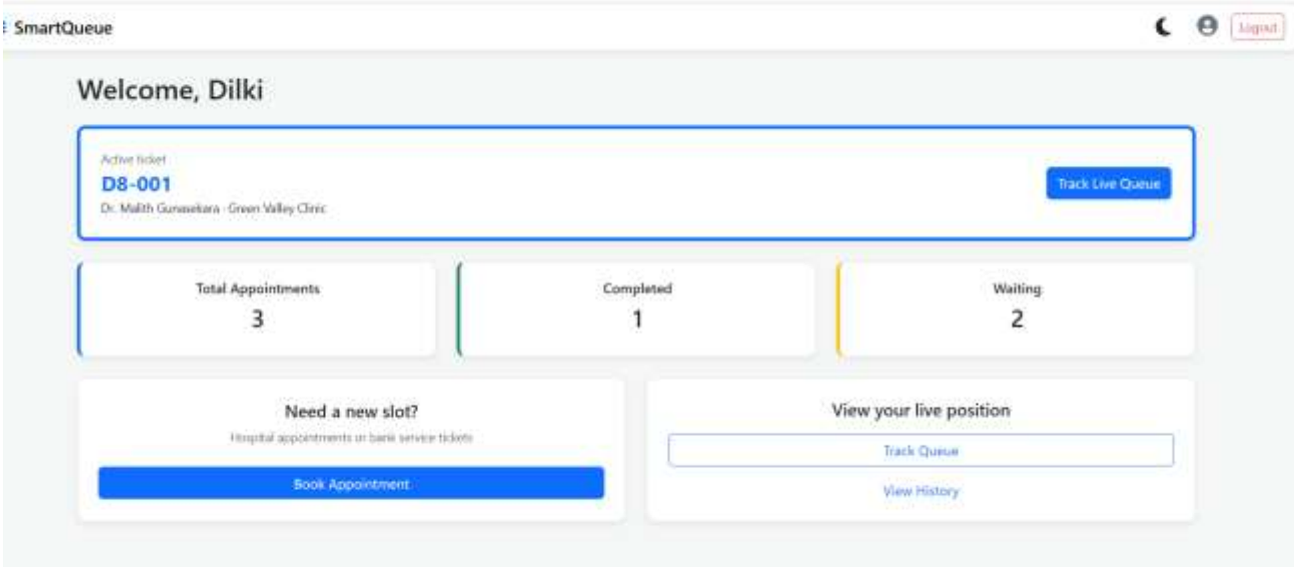
- Login Page



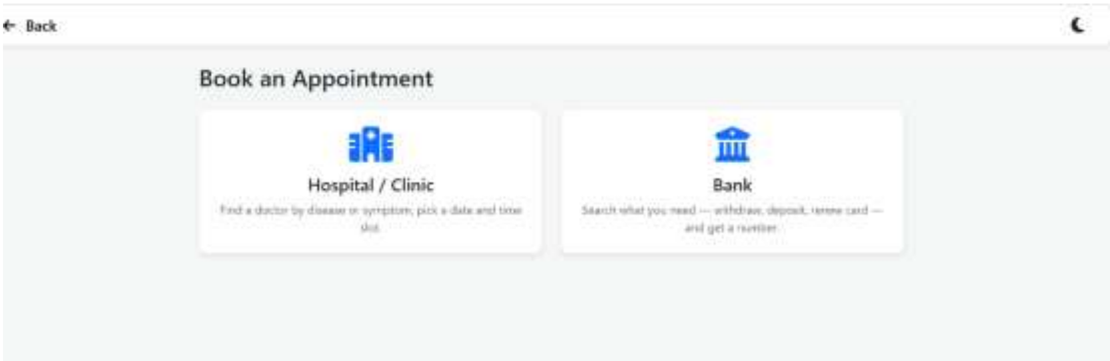
- Registration Page

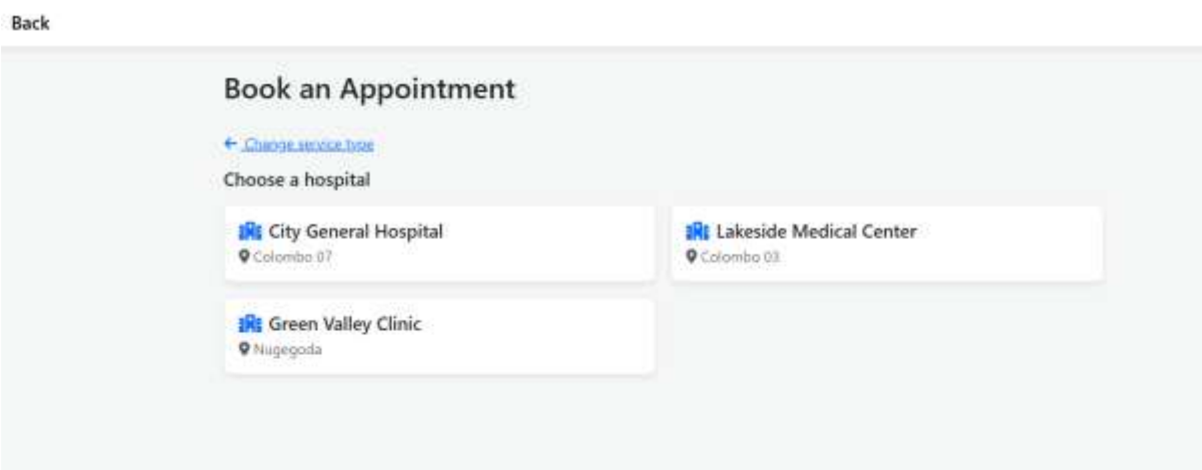


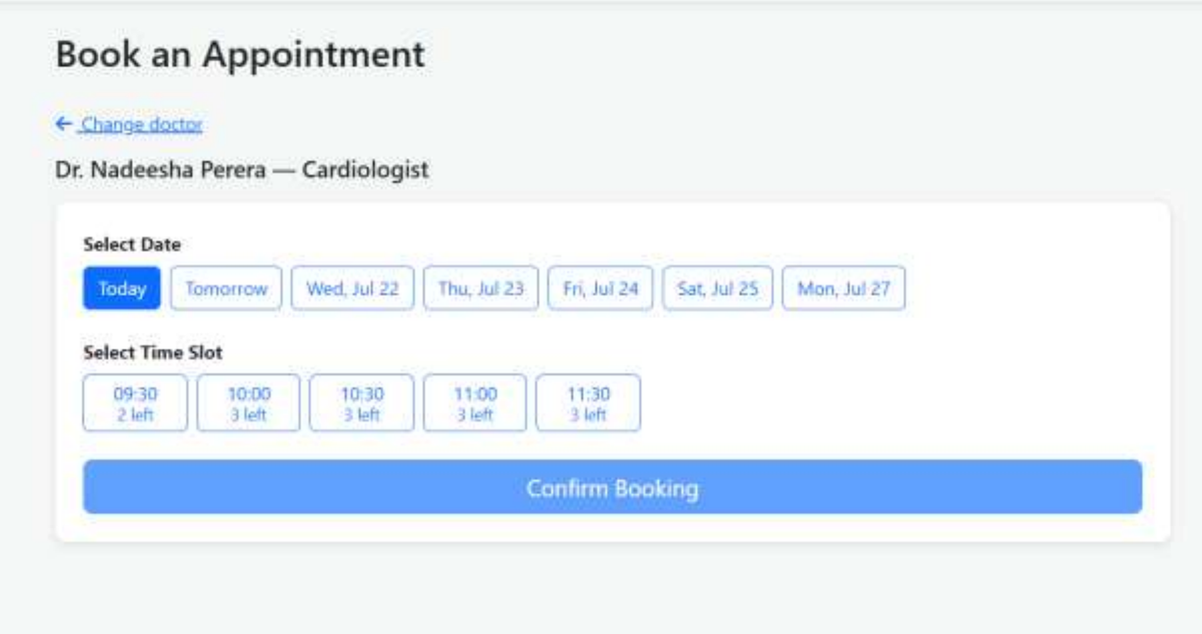
- Dashboard

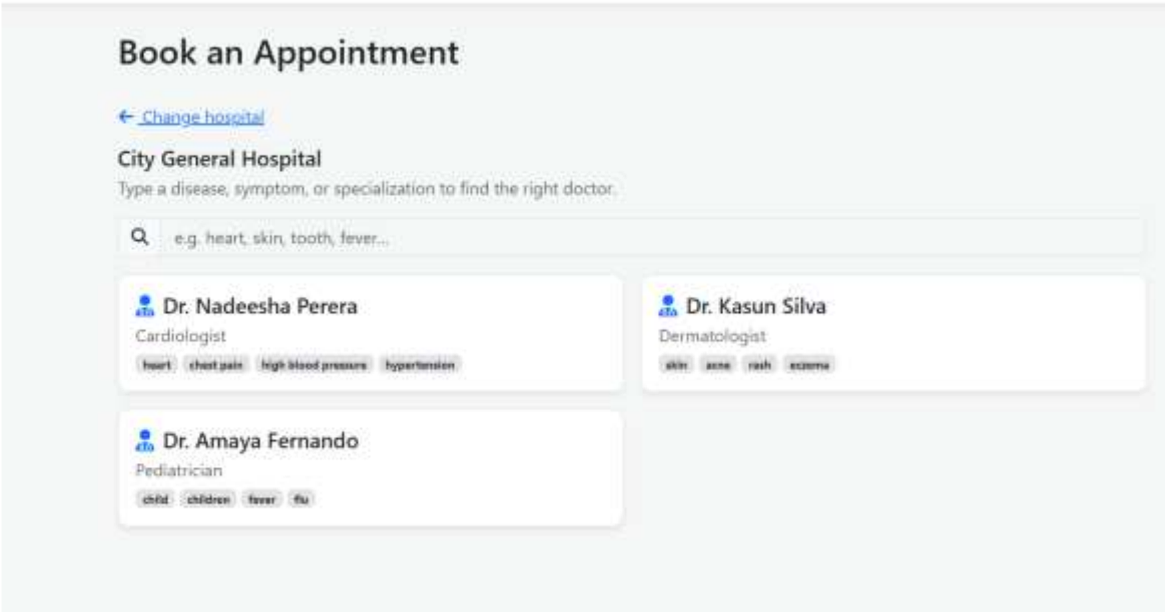


- Appointment Booking

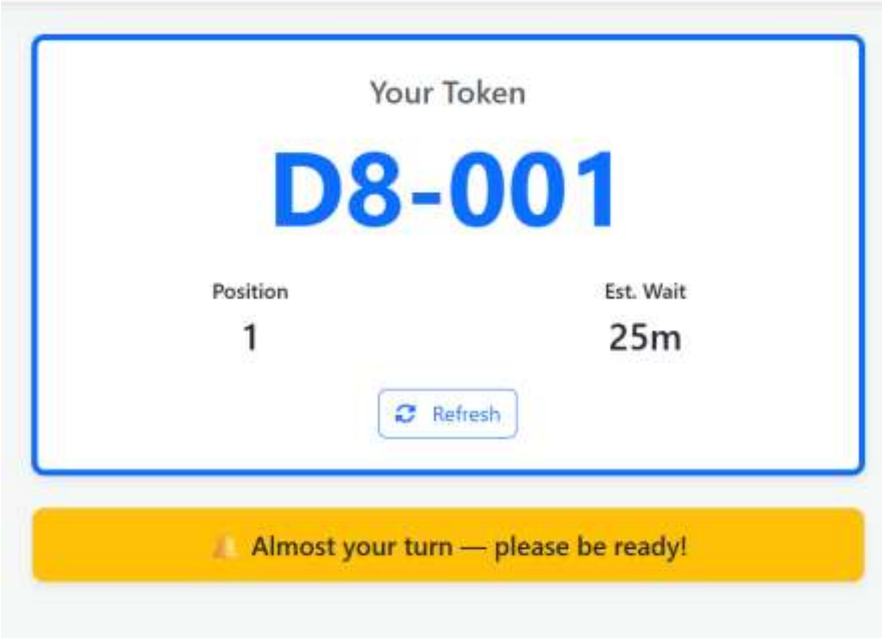




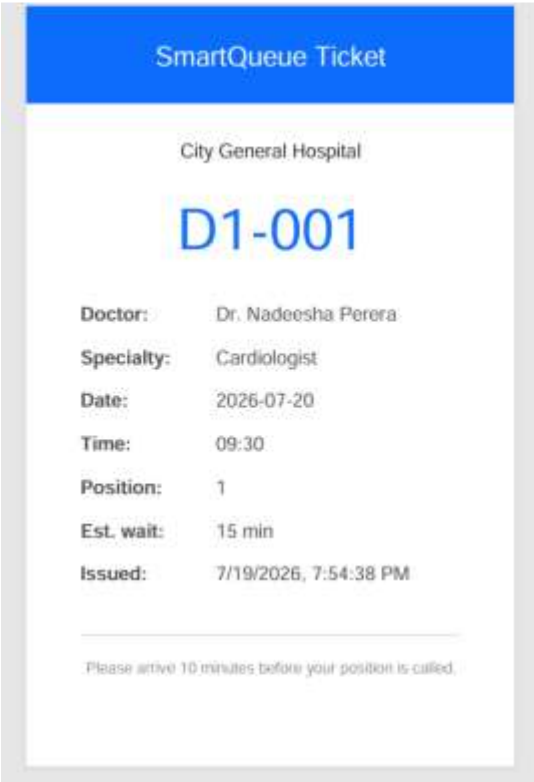

Book an Appointment
← Change doctor
Dr. Nadeesha Perera — Cardiologist
Select Date
Today
Tomorrow
Wed, Jul 22
Thu, Jul 23
Fri, Jul 24
Sat, Jul 25
Mon, Jul 27
Select Time Slot
09:30 2 left
10:00 3 left
10:30 3 left
11:00 3 left
11:30 3 left
Confirm Booking


Book an Appointment
← Change hospital
City General Hospital
Type a disease, symptom, or specialization to find the right doctor.
e.g. heart, skin, tooth, fever...
Dr. Nadeesha Perera
Cardiologist
heart
chest pain
high blood pressure
hypertension
Dr. Kasun Silva
Dermatologist
skin
acne
rash
eczema
Dr. Amaya Fernando
Pediatrician
child
children
fever
flu

- Queue Status

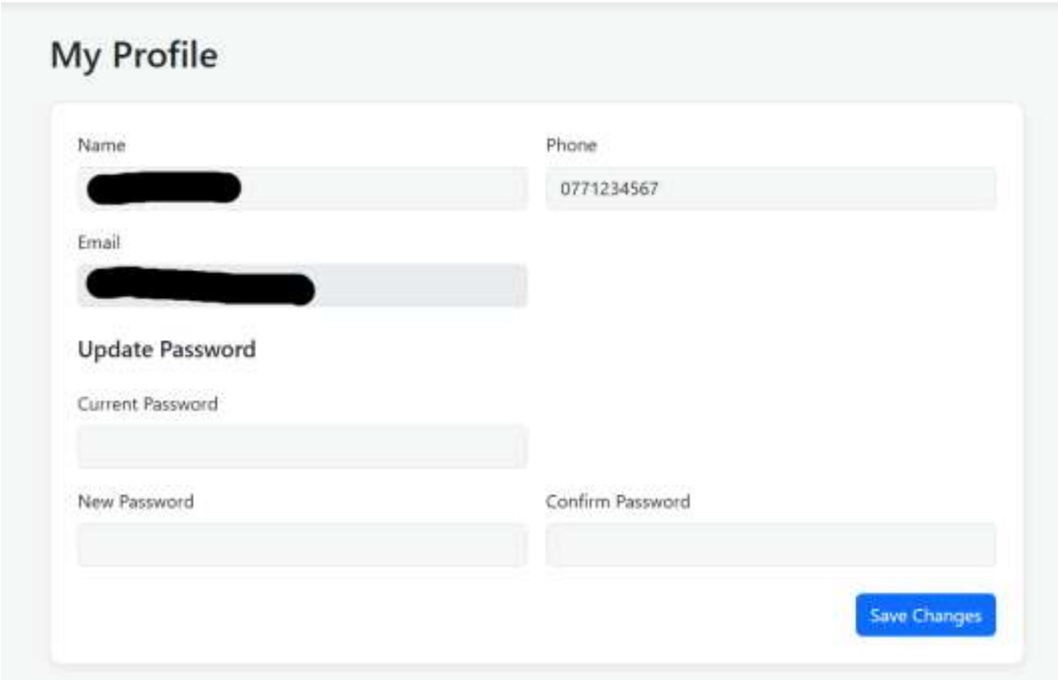


- Queue Ticket

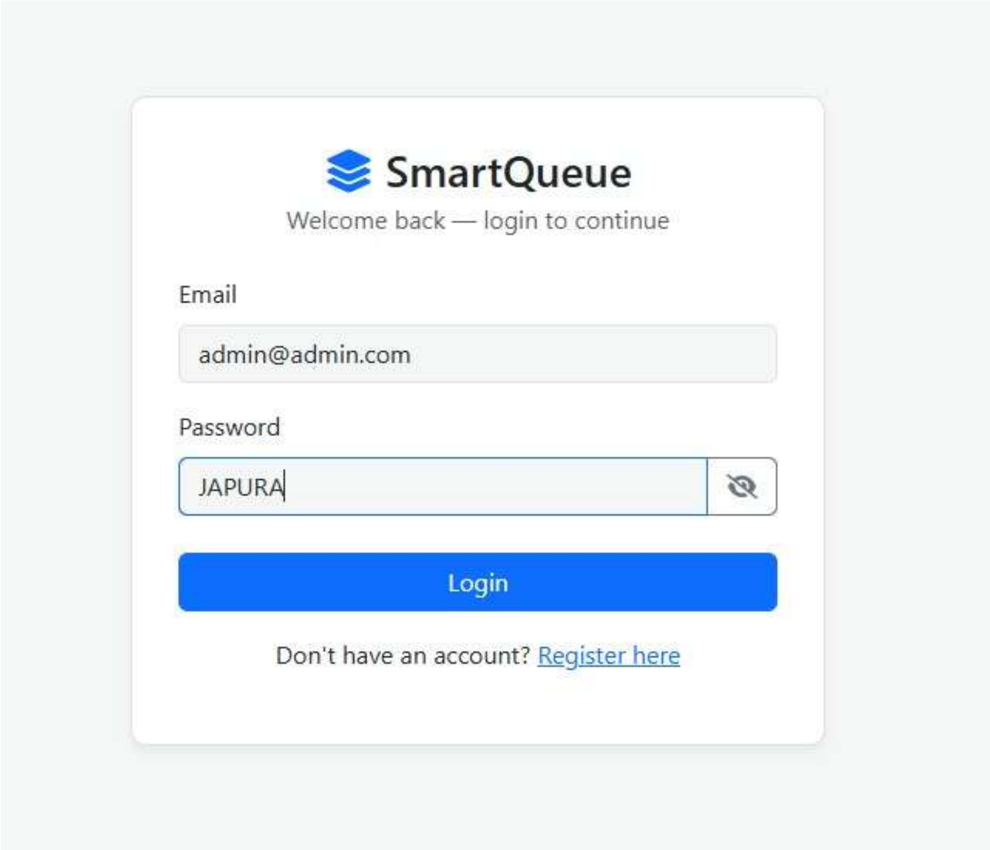


- Booking History

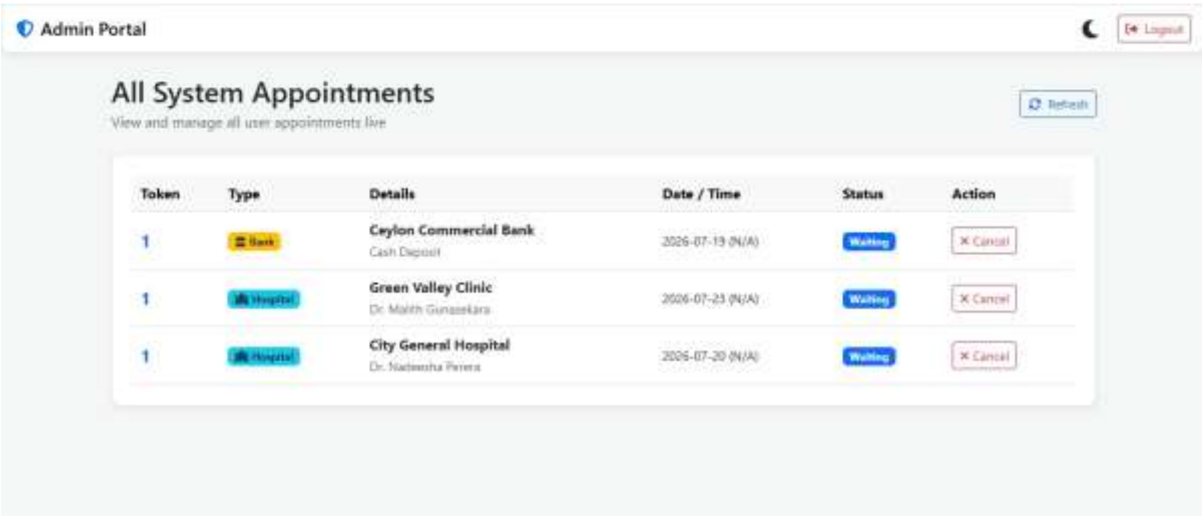
- User Profile


My Profile
Name
[redacted]
Phone
0771234567
Email
[redacted]
Update Password
Current Password
New Password
Confirm Password
Save Changes

- Admin Login



- Admin Portal



11. Testing

Test Case	Expected Result	Status
User Registration	New account created	Passed
User Login	Login successful	Passed
Book Appointment	Appointment created	Passed
Queue Generation	Queue number assigned	Passed
View Queue	Queue displayed	Passed
View History	Booking history shown	Passed
Logout	Session terminated	Passed

12. Future Improvements

Future enhancements include:

- MySQL database integration
- Java Spring Boot backend
- Email notifications
- SMS reminders
- QR code tickets
- AI-based waiting time prediction
- Mobile application (Android & iOS)
- Online payment integration
- Admin analytics dashboard
- Real-time queue updates using WebSockets

13. Conclusion

The Smart Queue Appointment System successfully addresses the inefficiencies of traditional queue management by providing a web-based digital solution. Users can conveniently register, book appointments, receive queue numbers, monitor queue status, and access booking history. The project demonstrates the effective use of Object-Oriented Programming principles, modern web technologies, and modular system design, making it scalable for future enhancements and suitable for real-world deployment.

14. References

1. Oracle Java Documentation
2. MDN Web Docs (HTML, CSS, JavaScript)
3. Node.js Documentation
4. GitHub Documentation
5. REST API Design Guidelines